



Technical Service Bulletin

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ISX15 and QSX15 Wide Valve Camshaft Lobe and Valve Rocker Lever Roller Release

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Core Issue

To improve the robustness of the ISX15 and QSX15 valve camshaft, the camshaft lobe and rocker lever rollers were widened. Increasing the width of these two components reduces the contact stress between the camshaft and the rocker lever rollers.

Confirmation

A compatibility matrix **must** be used to determine whether non-malfunctioning parts **must** be replaced. This matrix is shown in the camshaft and rocker lever Clean and Inspect for Reuse sections of the engine service manuals. The matrix is shown below.

Resolution

These new parts are **not** fully backward compatible to the previous versions of the valve camshaft and valve rocker lever roller.

Table 1, Parts Description

Part Description	Part Width	Part Width Measurement	Existing Part Number(s)	Obsolete	Superseded	New Part Number
Camshaft	Wide	20 mm [0.79 in]	4298626	No	No	-
Camshaft	Wide	20 mm [0.79 in]	4298627	No	No	-
Camshaft	Wide	20 mm [0.79 in]	4298628	No	No	-
Camshaft	Wide	20 mm [0.79 in]	4298629	No	No	-
Front Exhaust Rocker Lever	Wide	17 mm [0.67 in]	4386045, 4318204	Yes	Yes	5484228

Table 1, Parts Description						
Part Description	Part Width	Part Width Measurement	Existing Part Number(s)	Obsolete	Superseded	New Part Number
Rear Exhaust Rocker Lever	Wide	17 mm [0.67 in]	4386046, 4318205	Yes	Yes	5484229
Front Intake Rocker Lever	Wide	17 mm [0.67 in]	4386047, 4318206	Yes	Yes	5484230
Rear Intake Rocker Lever	Wide	17 mm [0.67 in]	4386048, 4318207	Yes	Yes	5484231
Camshaft	Narrow	18.5 mm [0.73 in]	4059331	Yes	Yes	4298626
Camshaft	Narrow	18.5 mm [0.73 in]	4059333	Yes	Yes	4298627
Camshaft	Narrow	18.5 mm [0.73 in]	3685963	Yes	Yes	4298628
Camshaft	Narrow	18.5 mm [0.73 in]	3685964	Yes	Yes	4298629
Front Exhaust Rocker Lever	Narrow	15 mm [0.59 in]	4972816, 4386045, 4318204	Yes	Yes	5484228
Rear Exhaust Rocker Lever	Narrow	15 mm [0.59 in]	4972817, 4386046, 4318205	Yes	Yes	5484229
Front Intake Rocker Lever	Narrow	15 mm [0.59 in]	4972818, 4386047, 4318206	Yes	Yes	5484230
Rear Intake Rocker Lever	Narrow	15 mm [0.59 in]	4972819, 4386048, 4318207	Yes	Yes	5484231

Table 2, Compatibility Matrix		
Part Description	Wide Camshaft Lobe	Narrow Camshaft Lobe
Wide Rocker Lever Roller	Yes	No
Narrow Rocker Lever Roller	Yes	Yes

Note : These design changes do **not** affect the maintenance intervals described in the Operation and Maintenance Manuals or Owner's Manuals for these engines.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2012-2-21	Module Created
2016-9-6	Modified rocker lever item description and updated wide part numbers in Table 1.
2018-4-27	Updated table 1.

Last Modified: 30-Apr-2018
